

block, these two interfaces will be imaged. In either case, some of the ultrasound will be absorbed and scattered as it passes through the silicon.

The amplitude of any return echo signal depends on the properties of the two materials at a given interface. The amplitude of an echo from the interface between two solids can range from just above a small percentage of the pulse to above 90 per cent, as seen in the centre of Fig. 1. But the amplitude of a return echo from any solid-to-air interface is invariably very nearly 100 per cent, as seen at right in Fig. 1. Even if the gap containing the air or another gas is as thin as a small fraction of a micron, the amplitude will be the same.

The time of flight of the return echo signals can also be recorded and used to measure the distance from a reference point to the interface from which the pulse is being reflected at a given x-y coordinate. In this way a buried contour such as a warped layer can be measured and imaged.

Fig. 2 shows the acoustic image of bonding of a heat-sink. The transducer scanned the area of the heat-sink and collected return echo signals from the gate depth, which encompassed the solder layer bonding the heat-sink to its substrate as well as the two interfaces (solder to heat-sink, and solder to substrate). Gray regions in the acoustic image represent partial reflection of the ultrasonic pulse from one or both of these interfaces.

Black spots are images of holes for bolts that will hold the heat-sink in place. These are black because

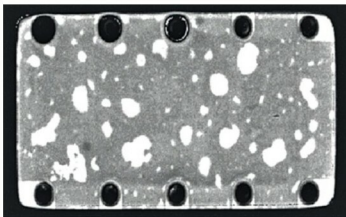


Fig. 2: White regions are air-filled heat-blocking voids in the adhesive material for this heat-sink

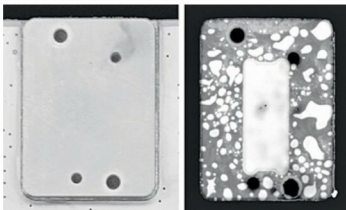


Fig. 3: Optical (left) and acoustic (right) images of a heat-sink on the back of a printed circuit board

they run completely through the gated depth—that is, there is no interface within the gated depth to reflect ultrasound.

The irregularly shaped white features are voids within the solder. These were formed by air bubbles trapped in the fluid solder. The voids block heat flowing from the die, as do the intended voids around the bolts. The solid-to-air interface at the face of the void nearest to the transducer reflects virtually all of the ultrasound, and thus appears white in the acoustic image. At three of the four corners of the heat-sink, what appears to be an intended void surrounds the bolt; solder flowed into the void at upper left. There are smaller intended voids that partly surround most of the other bolts.

An engineer examining this image would find plenty of information to use in process control.

In Fig. 3, on the left is the optical image of a heat-sink bolted to the back side of a printed circuit board, where it dissipates heat from a component on the top side of the board. The acoustic image of the same heat-sink is shown on the right.

The return echoes were gated on a depth from just above to just below the solder layer. The bolts are therefore black, similar to the bolts in Fig. 2. At the centre is the acoustic image of the cavity on the far side that holds the component. The numerous irregular white features are voids—air trapped in the fluid solder when the heat-sink was put in place. X-ray tends to miss thin features such as these because so little

of the solid material is absent.

Practically, any heat-sink that is joined to its substrate by solder or another adhesive is subject to voids, non-bonds and other gaps that block heat flow from the die. In addition to imaging the adhesive layer, an acoustic microscope can measure and report the percentage of the bond area that is occupied by gaps.

IGBT modules are more likely to be imaged than many other items because their high power levels and the types of applications (subway cars and mine drilling equipment, for example) make the presence of gaps that can impede heat flow more critical. An inverted acoustic micro imaging tool meets the requirement of imaging IGBT modules through the heat-sink. Its transducer faces up to pulse ultrasound into the module